

Extracting Safety Intelligence from Healthcare Incident Reports

Responsible AI in Healthcare

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About Me



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<https://cosmos-ai.com/>

Healthcare AI research spanning:

- Clinical text & reports
- Electronic health records
 - Physiological sensors
- Multimodal data fusion
- Temporal data modelling

Our Partner: HSIB → HSSIB

Healthcare Safety Investigation Branch (HSIB)

Established in 2017 to improve patient safety through independent investigations. We collaborated with HSIB on this research, analysing their maternity investigation reports.

Transformation in October 2023:

HSIB transformed into two separate organisations:

- **MNSI** — Maternity and Newborn Safety Investigations
- **HSSIB** — Health Services Safety Investigations Body

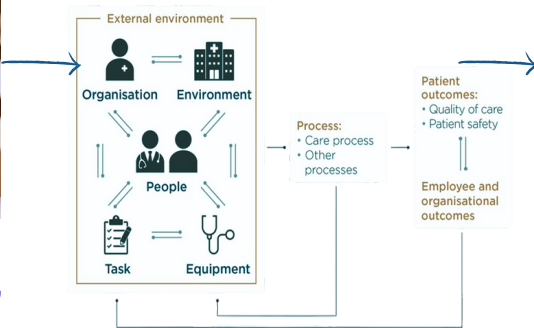
Core functions:

- Conduct independent patient safety investigations across NHS England
- Publish reports with safety recommendations and learnings
- Deliver patient safety investigation training

Investigations focus on **learning, not blame**. Staff who contribute are legally protected.

Extracting Insights from Maternity Incident Investigation Reports

A faster and more efficient way is needed for extracting insights from maternity incident investigation reports.



Supplementary File SF1. SIRch taxonomy of human factors for maternity investigations

- | | |
|---|---|
| <ol style="list-style-type: none"> 1. External Environment <ol style="list-style-type: none"> (a) Policy factor (b) Societal factor (c) Economic factor (d) COVID ✓ (e) Geographical factor (e.g. Location of patient) 2. Internal Environment <ol style="list-style-type: none"> (a) Physical layout and Environment (b) Acuity (e.g., capacity of the maternity unit as a whole) (c) Availability (e.g., operating theatres) (d) Time of day (e.g., night working or day of the week) 3. Organisation <ol style="list-style-type: none"> (a) Team culture factor (e.g., patient safety culture) (b) Incentive factor (e.g., performance evaluation) (c) Teamworking (d) Communication factor <ol style="list-style-type: none"> i. Between staff ii. Between staff and patient (verbal) (e) Documentation (f) Escalation/referral factor (including fresh eyes reviews) (g) National and/or local guidance (h) Language barrier 4. Jobs/Task <ol style="list-style-type: none"> (a) Assessment, investigation, testing, screening (e.g., holistic review) (b) Care planning (c) Dispensing, administering (d) Monitoring (e) Risk assessment (f) Situation awareness (e.g., loss of helicopter view) (g) Obstetric review 5. Technologies and Tools <ol style="list-style-type: none"> (a) Issues (b) Interpretation (e.g., CTG) 6. Person <ol style="list-style-type: none"> (a) Patient (characteristics and performance) | <ol style="list-style-type: none"> i. Characteristics <ol style="list-style-type: none"> A. Physical characteristics B. Psychological characteristics (e.g., stress, mental health) C. Language competence (English) D. Disability (e.g., hearing problems) E. Training and education (e.g., attendance at ante-natal classes) F. Record of attendance (e.g., failure to attend antenatal classes) ii. Performance <ol style="list-style-type: none"> A. Slip or lapse (errors that tend to happen in routine tasks that people are doing without much conscious thought) B. Decision error (errors in conscious judgements, decisions due to lack of knowledge and from misunderstanding of a situation) C. Intentional rule breaking (deliberately do something different from rules) (b) Staff (characteristics and performance) <ol style="list-style-type: none"> i. Characteristics <ol style="list-style-type: none"> A. Physical characteristics B. Psychological characteristics (e.g., stress, mental health) C. Language competence (English) D. Disability (e.g., hearing problems) E. Training and education (e.g., attendance at ante-natal classes) F. Record of attendance (e.g., failure to attend antenatal classes) ii. Performance <ol style="list-style-type: none"> A. Slip or lapse (errors that tend to happen in routine tasks that people are doing without much conscious thought) B. Decision error (errors in conscious judgements, decisions due to lack of knowledge and from misunderstanding of a situation) C. Intentional rule breaking (deliberately do something different from rules) |
|---|---|

What We'll Explore Today

1. **The Challenge:** how can AI help extract, organise, and analyse important findings from incident investigation reports?
2. **The Methods:** what AI techniques did we use? How did we develop and test them? Can AI find patterns reliably and fairly across all patient groups?
3. **The Ethics:** how did we ensure AI systems remain accountable? How did we keep humans in the loop and in control?

Using UK maternity safety as our case study

The Challenge

Making sense of safety data at scale

The problem:

- Healthcare generates vast **unstructured text**
- Safety reports contain actionable insights
- Patterns are **hard to spot** at scale
- Manual analysis cannot keep pace

By identifying recurring themes, we can help prevent future harm.

The considerations:

- Health data requires careful handling under GDPR
- AI must perform **fairly across all groups**
- Findings must be **traceable** and verifiable
- Human oversight remains essential

Responsible AI means designing in safeguards from the start.

*How can we use AI to extract
actionable safety intelligence
from hundreds of
unstructured reports
— responsibly and fairly?*

01

The Methods

How we built and tested the AI models

What is Concept Annotation?

Extracting human factors from investigation text

Step 1 | Read the sentence

Investigation report:

"The **abnormal CTG** was not recognised and did not prompt **escalation to the obstetric team**, in line with **local guidance**."

Step 2 | Assign concept labels

Interpretation Escalation/referral Local guidance

*A single sentence can reveal **multiple** human factors. Systematic annotation lets us spot patterns across hundreds of reports.*

SEIPS Taxonomy (6 categories)

External Environment – Policy, Societal, Economic, COVID, Geographical

Internal Environment – Physical layout, Acuity, Availability, Time of day

Organisation – Teamworking, Communication, Escalation/referral, Guidance

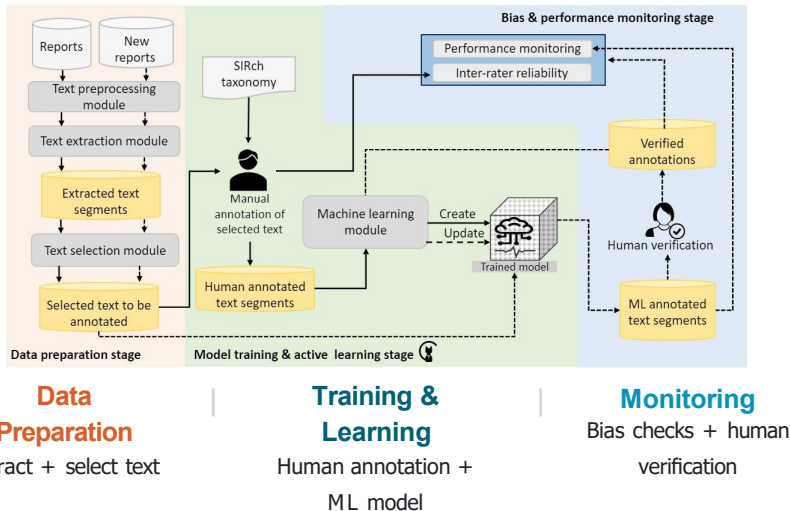
Job/Task – Assessment, Care planning, Monitoring, Risk assessment

Technologies & Tools – Issues, Interpretation (CTG)

Person – Physical, Psychological, Slips/lapses, Decision errors

I-SIRch: Framework Overview

End-to-end pipeline for safety intelligence extraction



Why not generic AI?

- ✓ Domain-specific
- ✓ Runs offline
- ✓ Fully traceable
- ✓ Human-in-the-loop

The SIRch Taxonomy: SEIPS Framework

Human factors framework for maternity safety investigations

1. External Environment (5)

Policy, Societal, Economic, COVID, Geographical

2. Internal Environment (4)

Physical layout, Acuity, Availability, Time of day

3. Organisation (8)

Team culture, Incentive, Teamworking,
Communication, Documentation,
Escalation/referral, National/local guidance,
Language barrier

4. Job/Task (7)

Assessment, Care planning, Dispensing,
Monitoring, Risk assessment, Situation awareness,
Obstetric review

5. Technologies & Tools (2)

Issues, Interpretation (e.g. CTG)

6. Person — Patient (9)

Physical, Psychological, Language, Disability,
Training, Attendance, Slip/lapse, Decision error,
Rule breaking

7. Person — Staff (9)

Physical, Psychological, Language, Disability,
Training, Attendance, Slip/lapse, Decision error,
Rule breaking

44 factors in full taxonomy.
27 appeared in our 188 maternity reports.

I-SIRch: Training the Model

Expert annotation with human-in-the-loop validation

Human annotation process:

1. Three domain experts annotated reports
2. Used SIRch taxonomy
3. **Ethnicity blinded** during annotation
4. Inter-rater reliability: **80.15%**

Blinding ethnicity during annotation prevents biases entering the training data.

Dataset composition:

188

Real investigation reports

2019–2022, 3,122 sentences

- Training: 91 reports (1,162 sentences)
- Testing: 97 reports (1,960 sentences)

Key principle: Train on real data, test on unseen data to validate real-world performance.

Dataset: Training and Testing Batches

188 maternity investigation reports (2019–2022) from HSIB

Batch	No. Reports	No. Concepts	No. Sentences	Purpose
Batch 1	76 real	970	818	Initial Training
Batch 2	15 real	452	344	Retraining
Batch 3	97 real	2644	1960	Testing (Real)
Batch 4	76 synthetic	970	818	Testing (Synth)

Ethnic Group Distribution (Batch 1):

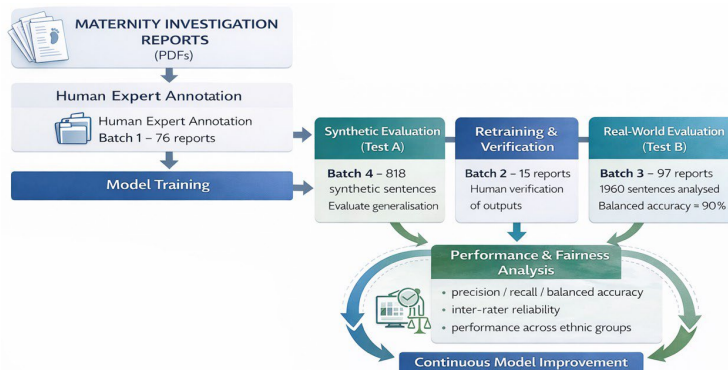
White British: 52 (70.93%) Asian: 6 (8.97%) Black: 7 (8.35%)

White Other: 6 (4.74%) Mixed Background: 1 (1.34%)

Human Annotation: 3 experts (HSIB + Human Factors) Inter-rater reliability: **80.15%**

I-SIRch: Development & Evaluation Cycle

Rigorous iterative validation across multiple batches



Why synthetic data?

Batch 4 tests whether the model genuinely learned concepts or merely memorised patterns.

Continuous fairness evaluation:

Performance tested across ethnic groups at each stage

I-SIRch: Evaluation Approach

Rigorous testing to ensure reliability and fairness

Performance evaluation:

- ✓ Tested on real unseen data (97 reports)
- ✓ Tested on synthetic data to check generalisation
- ✓ Multiple metrics: balanced accuracy, recall, precision, F-score
- ✓ Iterative retraining and retesting cycle

Human validation:

- ✓ Three domain experts annotated reports
- ✓ Inter-rater reliability measured (80.15%)
- ✓ Human verification of AI outputs

Fairness evaluation:

- ✓ Performance tested across ethnic groups
- ✓ Statistical tests for group differences
- ✓ Ethnicity blinded during annotation

Bias mitigation:

- ✓ Continuous monitoring for performance drift
- ✓ Retraining capability as new data arrives
- ✓ Sentence-level traceability for verification

I-SIRch: Overall Performance

Tested on 97 real unseen reports

Metric	Score
Balanced Accuracy	0.90 ± 0.18
Recall	0.93 ± 0.18
Precision	0.87 ± 0.34
F-score	0.96 ± 0.10

Balanced Accuracy = Overall accuracy corrected for label imbalance

Recall = How many actual issues did the AI catch?

Precision = When the AI flags an issue, is it correct?

Key finding: High recall (93%) means the model catches the vast majority of safety issues.

Tested on real, unseen data to validate real-world performance.

But does it perform **equally** for all groups?

I-SIRch: Fairness Across Ethnic Groups

Does the model perform equally for all groups?

Performance by ethnicity:

Ethnic Group	Bal. Acc.	Recall	n
White British	84%	67%	688
Asian	84%	68%	87
Black	82%	65%	81
Mixed Background	85%	69%	13
Other White	82%	64%	46

n = number of concept annotations tested

Precision = 1.0 (std=0) across all groups

Fairness evaluated on synthetic data (ethnicity unavailable for real test set)

Statistical analysis:

No significant difference between Black and White British, and Asian and White groups.

The model performs **consistently** across all ethnic groups — no evidence of bias.

AI systems must be tested for fairness across protected characteristics. ✓

02

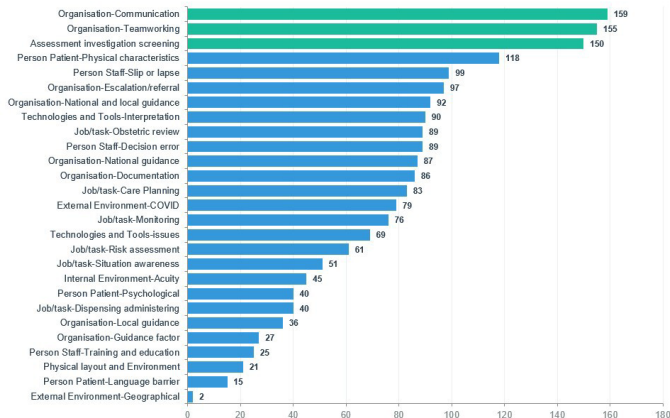
The Findings

What the AI revealed about maternity safety

*What patterns emerge
when AI analyses
188 maternity investigation reports
systematically and equitably?*

Human Factors Distribution Across Reports

188 reports, 3,122 annotated sentences



Source: 188 HSIB maternity investigation reports (2019-2022)

Adverse Outcomes:

- Therapeutic hypothermia
- Maternal death
- Stillbirth

The three green bars highlight the most common factors:

- **Communication:** 159 reports
- **Teamworking:** 155 reports
- **Assessment/investigation/screening:** 150 reports

Each bar shows the number of reports where that human factor appeared, e.g. Communication appeared in 159 out of 188 reports.

Communication

Not listened to, dismissing or ignoring women's concerns, preferences and requests

Not providing sufficient information for informed decisions

Conflicting recollections of discussions

Problematic staff interactions with women and their families

Lack of translation services prevented understanding of concerns and care options



Identified in **159 of 188** investigation reports

Teamworking

Fragmented triage care with many staff and inconsistent handovers

Communication and handover issues between teams

Hierarchy perceptions and communication gaps on needed interventions

Ambulance delays and excess workloads without adequate cover



Identified in **155 of 188** investigation reports

Assessment

Gaps in assessments and documentation, lack of key lab tests

Failure to recognise start of second stage of labour

Failures to appreciate full clinical pictures and perform key examinations

Busy units prevented appreciating mothers' concerns and performing needed assessments

Missed assessments, examinations, and reviews



Identified in **150 of 188** investigation reports

03

Summarisation

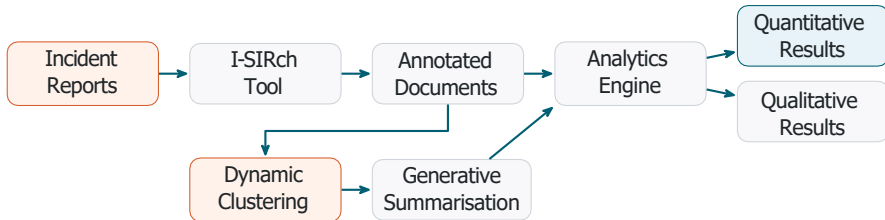
Multi-document synthesis by human factor

*Can we automatically synthesise
findings across 188 reports
into coherent summaries
for each human factor?*



Method 2: Multi-Document Summarisation

I-SIRch:CS Framework



Technical approach:

Sentence embeddings (all-MiniLM-L6-v2) →
K-means clustering → Abstractive
summarisation with **BART**

Offline models ensure GDPR compliance

Best model (BART) performance:

Relevance: **0.922** | Coherence: **0.794** |
Diversity: **0.806**

Dataset: 188 reports, 3,760 annotated
sentences across 27 concepts

Summarisation Dataset Composition

3,760 annotated sentences across 27 human factors concepts

ID	Concept	Index	Label
1	Acuity (54)	2	Antenatal (69)
3	Assessment, investigation, testing, screening (381)	4	COVID (142)
5	Care Planning (132)	6	Communication (477)
7	Decision error (169)	8	Dispensing, administering (62)
9	Documentation (168)	10	Escalation/referral factor (158)
11	Guidance factor (42)	12	Interpretation (197)
13	Language barrier (30)	14	Local guidance (88)
15	Monitoring (118)	16	National and local guidance (80)
17	National guidance (169)	18	Obstetric review (147)
19	Physical characteristics (320)	20	Physical layout and Environment (35)
21	Psychological characteristics (54)	22	Risk assessment (94)
23	Situation awareness (77)	24	Slip or lapse (188)
25	Teamworking (530)	26	Technologies and Tools-issues (112)
27	Training and education (40)		

The numbers in brackets are total annotations — how many sentences were tagged with each factor across all reports. E.g. Communication had 477 annotated sentences, Teamworking had 530.

Evaluating Equity in Summarisation

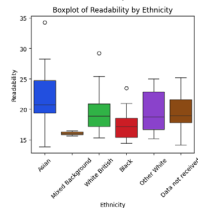
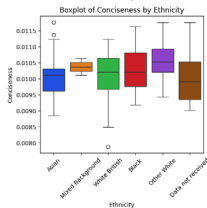
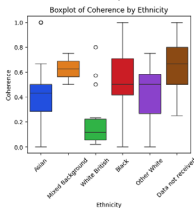
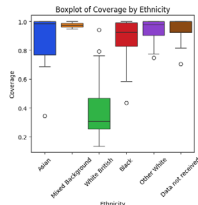
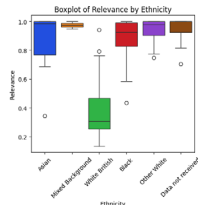
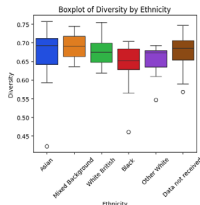
Do summaries perform fairly across ethnic groups?

BART performance by ethnicity:

Group	Rel.	Coh.	Div.
Asian	0.89	0.46	0.67
Black	0.87	0.54	0.64
Mixed	0.97	0.63	0.69
Other White	0.93	0.42	0.65
White British	0.40	0.18	0.68

Ethnicity data: 76 of 188 reports (Batch 1)

Lower White British scores attributed to
larger sample size introducing variability,
not model bias.



Fairness, transparency, and accountability must be designed in from the start.

How We Evaluate Summary Quality

Six metrics assess different quality dimensions

- **Diversity:** Variety of vocabulary (unique words / total words)
- **Relevance:** Similarity to original text (cosine similarity)
- **Coverage:** Proportion of key concepts captured
- **Coherence:** Logical flow between sentences
- **Conciseness:** Brevity of summary
- **Readability:** Ease of understanding (Flesch score)

Higher values indicate better performance.

These metrics enable objective comparison across models and demographic groups

Example Summaries Generated by BART

Demonstrating summarisation quality and traceability

Communication concept:

Heading: Staff not heard

Staff voiced concerns to the senior obstetrician but were left feeling unheard. No formal debriefing occurred afterwards, which staff valued. The opportunity to share reflections wasn't completed with all involved. Incremental delays and communication breakdown further impacted the DDI.

Acuity concept:

Heading: Reviews with seniors did not occur

The ambulance Trust experienced high call volumes. Due to high labour ward acuity, initial decisions weren't discussed with senior clinicians. Senior face-to-face review didn't occur until 16:05, 2 hours 50 minutes after recognising abnormalities. The abnormal CTG wasn't reviewed by the senior obstetrician.

Each summary is traceable to source file and sentence IDs, enabling verification and accountability.

Traceability in Summarisation

Linking summary content back to source documents for transparency and auditability

Ethnic Group	Outcome	Summary	Related IDs	Sentence
Asian	Cooled baby	High workload prevented timely review of abnormal CTGs and recognition of risk factors, delaying intervention and impacting cooled baby outcomes.]	370, 373, 374	
Asian	Early NND	Delays in post-delivery assessments of a baby prevented recognition of condition and need for monitoring, impacting an early NND.		372
Asian	Stillbirth	Incomplete maternal observations and assessments contributed to lack of recognition of clinical deterioration prior to stillbirths.	366, 367, 368	
Black	Cooled baby	Lack of holistic assessments meant missed screening for gestational diabetes and growth scans, potentially impacting cooled baby outcomes.	289, 290, 291, 292	
Black	MD	Lack of pregnancy test and referral to additional support represented missed care opportunities prior to a maternal death.		377, 378
Data not received	MD	Gaps in assessments and documentation, and lack of key lab tests represented missed opportunities prior to maternal deaths.	379, 380, 381	
Mixed Background	Cooled baby	Lack of testing for Zika virus prevented full risk assessment for a cooled baby.		371

Ethical Risks of AI Summarisation

Identified risks and mitigations

Risk	Mitigation
Information hallucination & bias amplification	Diversify training datasets; implement debiasing techniques; cross-reference with original texts; regular bias audits
Inadequate content control	Develop models with embedded ethical constraints; employ sentence ID traceability for verification
Sensitive data exposure	Use offline LLMs to reduce breach risks; balance security against performance needs
Sample size variability	Robust data preprocessing; ML algorithms customised for high variability datasets

Key safeguard: Sentence-level traceability enables verification of every summary, ensuring verifiability, trustworthiness, and accountability.

*How we applied this
framework work for other
types of investigations?*



04

LENS.ai

Learning Through Evidence from Natural Language Systems

Beyond Maternity: LENS.ai



Extending the framework to Prevention of Future Death reports

What are PFD reports?

Reports issued by coroners after inquests where action may help prevent future deaths. They highlight risks, system failures, and opportunities for learning across organisations.

What LENS.ai supports

- Systematic analysis of coronial reports and responses
- Identification of recurring risks, themes, and contributing factors
- Structured analysis across cases, years, factors, combinations
- Adaptable to different analytical frameworks (e.g. Yorkshire Contributory Factors)

REGULATION 28: REPORT TO PREVENT FUTURE DEATHS

REGULATION 28 REPORT TO PREVENT FUTURE DEATHS THIS REPORT IS BEING SENT TO: 1. Alex Whitfield Chief Executive Hampshire Hospitals NHS Foundation Trust Royal Hampshire County Hospital, Romsey Road Winchester, Hampshire, SO22 5DG 2. [REDACTED] Park & St Francis Surgery Hursley Road, Chandlers Ford, Eastleigh, HantsSO53 2ZH	
1	CORONER I am Karen Harrold, Assistant Coroner for the coroner area of Central Hampshire.
2	CORONER'S LEGAL POWERS I make this report under paragraph 7, Schedule 5, of the Coroners and Justice Act 2009 and regulations 28 and 29 of the Coroners (Investigations) Regulations 2013. http://www.legislation.gov.uk/ukpga/2009/7/schedule/5 http://www.legislation.gov.uk/ukpga/2013/1629/made
3	INVESTIGATION and INQUEST On 19 th June 2017, the Senior Coroner, Grahame Short, commenced an investigation into the death of Joan Elizabeth Betteridge aged 88 years old. The investigation concluded at the end of the inquest on 13 th December 2017. I recorded a Narrative Conclusion as follows: Joan Elizabeth Betteridge had a history of falls including the 31st May 2017. She was admitted to the Royal Hampshire County Hospital in Winchester on 9th June 2017. An X-ray confirmed a displaced fractured neck of left femur requiring surgery namely cemented hemiarthroplasty which took place on 10th June. Mrs Betteridge remained stable for 50-55 minutes into the operation and 3 minutes after cement insertion she developed a slow heart rate which despite immediate treatment resulted in her death. The post mortem confirmed the cause of death was bone cement implantation syndrome, a recognised complication of this type of operation. The medical cause of death was recorded as: 1a) Bone cement implantation syndrome; 2) Ischaemic heart disease.

LENS.ai: The Platform

Learning Through Evidence from Natural Language Systems

Key capabilities

- **Search and exploration**

Browse reports using keywords and filters

- **Pattern discovery**

Identify trends across themes, time, and regions

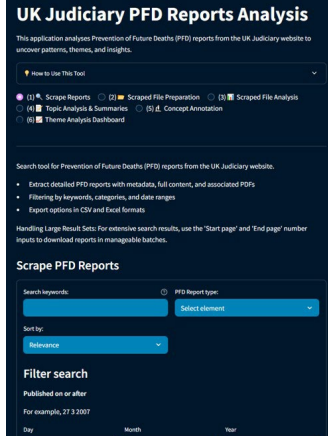
- **Thematic analysis**

Extract recurring concepts and contributing factors

- **Interactive outputs**

Visualise findings and export data

Designed for use beyond technical audiences



A web-based platform for exploring and analysing coronial safety data

LENS.ai: Annotation in Action

Colour-coded themes with full traceability

Highlighted Text				
are as follows. Evidence was given at the request and it is accepted, that many of the concerns identified in relation to Eciqur's birth have now been addressed, but some remain. 1. [Risk analysis / mode and timing of birth: A significant factor was the failure to consider an elective Caesarean Section (CS). It is not clear why on this history and consideration of mode of delivery did not include a consideration of an elective CS at any stage, by any of the doctors involved.] Evidence was given that existing policies in place at GCH were not followed and the reason for this failure has not been identified. Whilst Badgermet could be a useful tool, the evidence suggested that it is being regarded as a first line of defence rather than a failure. 2. I am told that the Badgermet maternity system implemented in October 2021 will resolve the problems relating to mode and timing of birth, in that the system will require a consultant to approve a decision for CS/induction of labour (IOL) and that IOL's will not be approved by the labour ward coordinator unless the paper booking has been approved on paper. However, on the evidence received it is not clear how this will assist if the risk factors and the need to consider IOL/elective CS have not been identified, by junior staff, thus triggering the need for escalation to approval by a consultant. 3. I am not clear what has been done to ensure that junior doctors and locums have sufficient technical knowledge to ensure that they are able to identify serious risk factors and alert the consultant to these, so that the consultant may consider the appropriate mode of delivery. 4. Information was provided as to the availability of regular face to face training for all grades concerning high-risk pregnancies not just for career trainees and foundation doctors. I am told that a memo was sent to all junior doctors reminding them that any plan for either IOL or elective CS must be approved by a consultant and that an induction pack containing that information is provided to new starters and locums. 5. I have seen the Women's and Children's Clinical Education Guidelines introduced in November 2022. 6. I am told that the information was placed in a prominent position on notice boards, staff rooms and in blogs. 7. It is not clear how the assurance of this knowledge is tested. 8. It was suggested that this may be in appraisal, but this seems only likely to identify problems after they arise. 9. In any event it seems these methods of dissemination of information and the appraisal system were in place at the time of Eciqur's birth but did not ensure that the doctors involved in her mother's care appreciated the impact of the risk factors in this pregnancy. 10. It was not explained how Badgermet or any policy or procedure in place, would ensure that there is a holistic review (including ultrasound scan findings) when planning for timing and mode of birth. The concern remains that there will be further failure to ensure that appropriate timing and mode of delivery will be provided in high-risk pregnancies. 11. Access to Badgermet portal / full implementation of the Badgermet software. It seems that the Badgermet system is being used on to address a number of issues which were identified in this case, and history reliance is being placed on a system which is not yet fully implemented. [The concern remains that a critical aspect of this system, access to the portal, will not be in place until autumn of 2023 at the earliest. 12. Quality Assurance. I am advised that Badgermet can easily identify faults which have not been completed and will prevent a record being closed until the fault is completed, but I cannot identify the quality of any such entries. 13. I am told that the quality checks are made on ten sets of notes per month out of an estimated 3,000 records that will be open at any one time. The concern remains that there is insufficient quality assurance in this system. 14. Procedure for escalating concerns arising out of Ultrasound Scans (USS). I understand that the procedure for escalating concerns arising out of a USS where it is not possible to obtain an accurate measurement in a high-risk pregnancy is currently under review. I am told that the new policy is not in place. The concern remains that no date has been set for its implementation. 15. Counselling for mothers at increased risk of shoulder dystocia have not seen any indication that all mothers identified to have an increased chance of shoulder dystocia now receive counselling regarding the risks and benefits associated with vaginal birth or CS. Assisting mothers to understand the implications of risks they face is fundamental to supporting them to make fully informed decisions, in accordance with Montgomery. [The concern remains that there is no clear plan in place to ensure mothers receive the support they require to make fully informed decisions in relation to mode of delivery where there is risk of shoulder dystocia. 16. Interpretation of Montgomery. Ayle-mat care is unique in that decisions have to be made for the benefit of two patients (the mother and the baby) and the treatment options for each may, as in this case have competing risks and benefits. In addition, in ante-natal care, the circumstances may change and action may need to be taken very quickly. On the basis of evidence given at the request, there seems to be a lack of clarity as to the way in which Montgomery guidelines are interpreted. It was acknowledged in evidence that parents often want a steer as to the best/ safest course of action and that may require medical professionals to express opinions as to the weight to be placed on increased risk factors. In some cases, parents may prefer to rely on the viewpoint of an experienced medical professional. It seems that medical professionals do not feel they can offer this assistance as it might be interpreted as trying to impose their opinion on the parent. The way in which Montgomery is interpreted and the extent to which medical professionals can offer an opinion is of wider concern than just the actions of those at GCH and should be considered by those who produce the guidance and deliver training to medical professionals.]				
Framework	Theme	Color	Matched Keywords	Extracted Sentence
Extended Analysis	Diagnostic Testing and Specimen Handling - I-Birth_Unternal - Geographical factor (e.g. Location of patient)	#DCECF1	analysis + factor	Risk analysis / mode and timing of birth: A significant factor was the failure to consider an elective Caesarean Section (CS).
House of Commons	Fragmented care	#FFCC00	clear	It is not clear why on this history and consideration of mode of delivery did not include a consideration of an elective CS at any stage, by any of the doctors involved.
House of Commons	Inadequate maternity care	#FF9999	maternity	I am told that the Badgermet maternity system implemented in October 2021 will resolve the problems relating to mode and timing of birth, in that the system will require a consultant to approve a decision for CS/induction of labour (IOL) and that IOL's will not be approved by the labour ward coordinator unless the paper booking has been approved on paper.

- Text is highlighted by theme

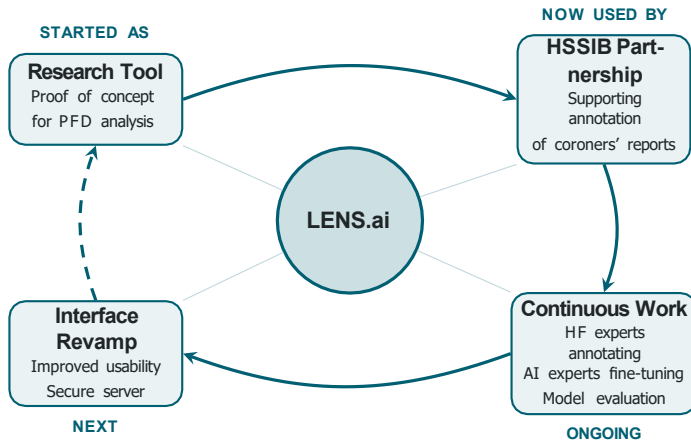
Fragmented care, guidance gaps, inadequate maternity care

- Each annotation links to the original sentence

Full traceability maintained

LENS.ai: A Living Platform

From research prototype to operational tool



Hundreds of reports annotated | Model fine-tuned per concept and process

Key Takeaways

Challenge

Unstructured safety data contains insights but is hard to analyse at scale.

Methods

Our tools combine human expertise with AI to extract safety intelligence fairly.

Ethics

Fairness, transparency, and human oversight must be designed in from the start.

AI assists, humans decide

Get in Touch

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Research website:
<https://cosmos-ai.com/>

If you are interested in:

- Finding out more about our work
- Using the tool to analyse your reports
- Research partnerships

The tool is designed to work both online and offline.

Thank You

Selected Publications

I-SIRch: AI-powered concept annotation tool for equitable extraction and analysis of safety insights from maternity investigations

International Journal of Population Data Science, 2024

doi: 10.23889/ijpds.v9i2.2439

Intelligent Multi-Document Summarisation for Extracting Insights on Racial Inequalities from Maternity Incident Investigation Reports

Artificial Intelligence in Healthcare (AIIH), Springer, 2024

<https://arxiv.org/abs/2407.08322>